Training-Free Partial-Order Control for Discrete-Diffusion POI Check-in Generation

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Abstract

Location-based social networks (LBSNs) are a core source of web-scale human-mobility data, yet sharing raw check-in traces is privacy-sensitive, motivating realistic synthetic generation. Beyond statistical realism, many downstream web applications need control over the *semantic order* of activities—e.g., leisure before dining—a sample-level, cross-position requirement that single-step conditioning cannot express. Enforcing such constraints is commonly assumed to cost distributional fidelity; we show this trade-off is largely *separable*. We present PCDG, a *training-free* control stack for a frozen conditional joint temporal + category/POI discrete-diffusion generator. PCDG compiles category partial-order constraints into differentiable energies, projects the category-position logits through a KL-preserving augmented-Lagrangian optimization in the late reverse steps, eliminates residual category–POI inconsistency via a category-consistent decoder, and restores spatial realism through a distribution-aware same-category geometry refinement that leaves every constraint metric unchanged *by construction*. Across two cities under an out-of-distribution split and three sampling seeds, PCDG cuts the strict ordering-violation rate from 92.89% to 4.04% on Istanbul and from 89.45% to 11.28% on New York, raising pairwise ordering coverage by 6.5× and 4.2×, while *improving* all six fidelity divergences over the unconstrained backbone on Istanbul and four of six on New York. On Istanbul it surpasses a *fully retrained* partial-order conditional model on all five constraint metrics and five of six fidelity divergences, without retraining anything. Stage-wise ablations localize where fidelity is lost and recovered across the three-stage stack.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Information systems** → *Information systems applications.*

Keywords

synthetic human mobility, discrete diffusion, partial-order constraints, training-free constrained sampling, POI check-in sequences

1 Introduction

Location-based social networks (LBSNs) produce time-stamped visits to points of interest (POIs), linking human mobility to activity categories such as dining, leisure, and shopping. These traces support POI recommendation, urban analytics, and simulation [9, 26, 41], but releasing raw traces exposes sensitive behavior. Synthetic mobility generation offers an alternative data source [7, 10, 24, 43].

Relevance to the Web. We study a control problem for LBSN data generation: how to generate complete check-in sequences under activity-order requirements while retaining the population statistics used in web-data analysis. Scenario-specific synthetic traces could support recommender stress tests and data augmentation without releasing individual logs. The evaluated contribution is controllable generation; downstream utility and formal privacy protection remain unevaluated.

A scenario may require leisure before dining or sightseeing before shopping. Such precedence couples multiple, non-adjacent positions. A local condition embedding does not ensure that every occurrence of one category precedes every occurrence of another, or that both categories appear. Reordering completed sequences can fix inversions but cannot supply missing activities. Road-topology control [46] addresses geographic structure, whereas constrained discrete diffusion (CDD) [3] provides a general framework for sampling under differentiable constraints. POI generation adds a further difficulty: category tokens and POI identities can disagree, so controlling the former need not control the actual activities of the emitted POIs.

We introduce PCDG, a training-free control stack on the frozen Marionette generator [7]. It adapts CDD's KL-anchored projection to sample-level order and existence energies, with a distance-distribution regularizer on POI logits. Category-consistent decoding then restricts POI support using the sampled category. After diffusion, same-category geometry refinement changes POI identities and coordinates while preserving actual categories, times, and event order. This separates spatial refinement from the fields that determine constraint satisfaction.

Across Istanbul and New York, the stack improves every constraint metric over the retrained PO-CFG baseline. Geometry refinement lowers spatial divergences without changing constraint metrics; the full stack improves all six fidelity JSDs over the unconstrained backbone on Istanbul and four on New York. New York CategoryTransition and G-RANK remain worse. These results identify a recoverable spatial cost in this generator, rather than establishing that every fidelity–control trade-off can be removed.

Contributions.

- We formulate category-precedence control for complete POI check-in sequences using order and existence energies (§3).
- We develop a training-free stack combining logit projection, category-consistent decoding, and constraint-neutral same-category geometry refinement (§4).
- We separate category-order control from spatial refinement and measure the resulting constraint–fidelity trade-offs, including residual category and popularity errors (§5.4).
- We provide a two-city, three-seed evaluation, projection ablations, and a stage-wise comparison under a fixed selection protocol (§5).

2 Related Work

Human mobility and POI modeling. Classical studies established that human movement is highly regular yet only boundedly predictable [13, 34]. A large body of work predicts the *next* location from history using attentional, spatio-temporal, and graph-enhanced models [9, 26, 41], and diffusion models have recently been applied to next-POI and sequential recommendation [6, 22, 29, 36, 38]. These methods target prediction or ranking of a single next step and do not emit a complete, constraint-satisfying sequence of time-stamped, categorized visits.

Synthetic trajectory and mobility generation. Data-driven generation spans GAN/Seq2Seq models [2, 19, 21, 40], point-process and neural-ODE generators [24, 43], imitation- and privacy-oriented approaches [10, 17, 28, 30, 35, 37], and, increasingly, diffusion models [4, 12, 23, 31, 39, 42, 45] and LLM-based generators [14, 16, 20, 44]. Closest to our backbone, Marionette performs fine-grained conditional generation of spatiotemporal trajectories with a diffusion point process and a category-to-POI cascade [7], and related benchmarks study imitative generation quality [8]. These works pursue distributional realism or coarse conditioning; none enforces a *sample-level partial order over activity categories* within the generated sequence.

Discrete diffusion. Discrete diffusion extends denoising diffusion [15] to categorical data via structured transition matrices [1], concrete score matching [27], score-entropy [25], and masked formulations [32]; guidance mechanisms steer discrete generation toward conditions [33]. We adopt a discrete-diffusion backbone and intervene at sampling time rather than changing training.

Controllable and constrained generation. ControlTraj injects road-topology constraints into trajectory diffusion [46]; Geo-Llama conditions an LLM on mobility constraints [20]. In NLP, structured decoding enforces grammars or DAG automata [5, 11], and precedence constraints are studied in temporal planning [18]. Most relevant methodologically, Constrained Discrete Diffusion (CDD) embeds differentiable constraint optimization into discrete-diffusion sampling via KL projection, Gumbel-softmax relaxation, and augmented Lagrangian updates [3]. PCDG *specializes* this sampling-time projection paradigm to the previously unaddressed case of activity-category precedence in joint temporal + category + POI check-in generation: we contribute the order/existence energy formulation for sample-level partial orders, a category-position-restricted projection coupled with a spatial distance-distribution regularizer on POI positions, and—unlike prior controllable generators that retrain or post-process—a training-free controller that we show outperforms a *fully retrained* partial-order conditional model on constraint satisfaction (§5).

3 Problem Formulation

POI check-in sequences. We model a user's daily behavior on an LBSN as a variable-length sequence of check-in events. A sequence of length n consists of arrival times $t_{1:n}$ (continuous, strictly increasing), activity categories $k_{1:n}$ with $k_i \in C = \{1, \ldots, C\}$, and points of interest $l_{1:n}$ with $l_i \in \mathcal{L}$. Each POI l carries fixed geographic coordinates $\mathrm{gps}(l)$, and each training POI is associated with a category. A per-sequence context $\mathbf{c}$ encodes coarse temporal and behavioral conditions (e.g., time-of-day occupancy indicators). The empirical distribution over such sequences is highly structured: it reflects the regularity and bounded predictability of human mobility [13, 34] while retaining substantial individual variability.

Category partial-order constraints. Many web applications do not merely require a statistically realistic sequence; they require that the *activity semantics* follow an application-specified order—for instance, that leisure precedes dining, or that sightseeing precedes shopping. We formalize such requirements as a sample-level *partial order* over categories. Given a reference or user-specified ordering, we encode it as a binary matrix $\mathbf{M} \in \{0,1\}^{C\times C}$, where $\mathbf{M}[a,b] = 1$ denotes the strict precedence constraint "category a precedes category b": in a satisfying sequence, *every* occurrence of category a must appear before *every* occurrence of category b. When $\mathbf{M}$ is derived from a reference sequence, we set $\mathbf{M}[a,b] = 1$ iff the last position of a precedes the first position of b. A constraint (a,b) is *violated* either by an inversion (some b before some a) or, degenerately, by dropping a or b entirely so that the ordering is vacuously "satisfied." A faithful method must avoid both.

Task. Given a frozen conditional generator of complete check-in sequences and a sample-level partial-order matrix $\mathbf{M}$, we seek to sample

$$(t_{1:n},\, k_{1:n},\, l_{1:n}) \sim p\big(t_{1:n}, k_{1:n}, l_{1:n} \mid \mathbf{c}, \mathbf{M}\big), \tag{1}$$

such that (i) the generated sequences remain close to the real data distribution in time, space, and activity semantics; and (ii) the category subsequence satisfies the precedence constraints encoded by $\mathbf{M}$. Unlike next-location prediction [9, 26, 41], the task outputs a *complete* time-stamped, categorized, geolocated sequence; unlike single-step conditioning, the constraint couples *multiple, non-adjacent* positions and therefore cannot be enforced by reweighting any single token independently.

Evaluation axes. We evaluate along two axes that are in tension. *Distributional fidelity* is measured by the Jensen–Shannon divergence (JSD) between generated and reference marginals of travel distance, radius of gyration, category-transition structure, daily location count, hourly category usage, and a global ranking statistic (G-RANK). *Constraint satisfaction* is measured by ordering-violation rates (OVR), pairwise and category coverage of the required relations, and an overall unsatisfied-sequence rate (Unsat); full definitions are given in Section 5. A trivial reordering can satisfy the constraint while destroying fidelity, and an unconstrained generator maximizes fidelity while ignoring the constraint; the goal is to improve constraint satisfaction with minimal fidelity cost.

4 Method

PCDG controls a frozen generator at sampling time through logit projection, category-consistent POI decoding, and terminal same-category geometry refinement. Figure 1 combines the framework and a controlled reverse step; ρ, λ, β enter the projection and refinement objectives, rather than forming a separate execution stage.

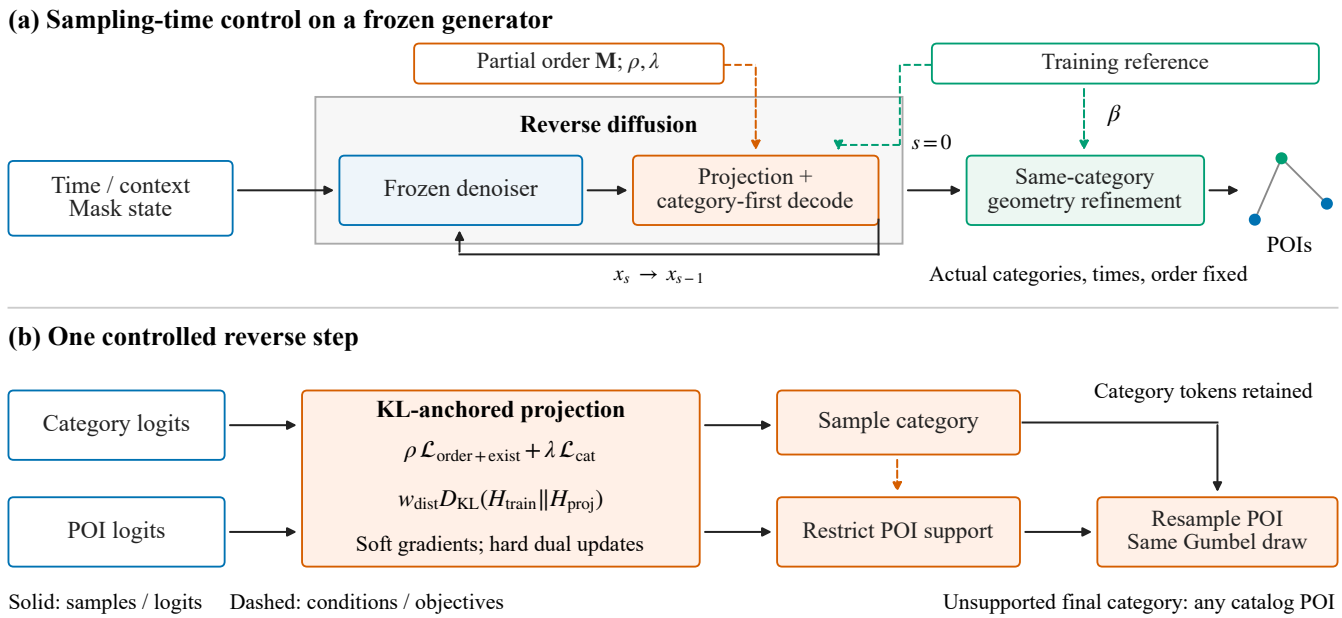


Figure 1: PCDG at sampling time. (a) Projection and decoding operate inside the frozen reverse process; same-category geometry refinement follows it, preserving actual categories, times, and event order. (b) Category energies and distance KL act on different logit positions. The decoder reuses the sampling noise and restricts POI support using the sampled category; unsupported final categories fall back to the catalog. Dashed edges carry conditions or objective inputs.

4.1 Backbone: conditional joint discrete diffusion

We build on a conditional generator that models arrival times with a diffusion point process and the discrete content (categories and POIs) with a conditional discrete diffusion model, following the fine-grained trajectory generator of Deng et al. [7]. The temporal branch samples event times; the discrete branch represents a sequence as a token string $\mathbf{x} = [\textsc{start}, k_{1:n}, \textsc{sep}, l_{1:n}, \textsc{end}]$, with a *category mask* and a *POI mask* identifying the two segments. Categories and POIs use separate absorbing-state transition matrices [1]; a Transformer denoiser with an $\mathbf{x}_0$-parameterization predicts, at each reverse step, per-position logits $\mathbf{z} \in \mathbb{R}^{V\times L}$ over the joint vocabulary of size V across the L positions, from which the posterior $p(\mathbf{x}_{s-1} \mid \mathbf{x}_s)$ and the next discrete state are sampled. **The backbone is used as-is and is never retrained by PCDG**; our contribution modifies only the per-step sampling distribution $\mathbf{z}$.

4.2 Partial-order constraints as differentiable energies

Given the sample-level matrix $\mathbf{M}$, we parse it into a set of pairwise constraints $\{(a,b) : \mathbf{M}[a,b] > \frac{1}{2}\}$. To make a constraint differentiable in the logits, we relax the hard categorical choice at each position into a distribution. With temperature T and (optionally) Gumbel noise $\mathbf{g}_l$,

$$\boldsymbol{\pi}_l = \mathrm{softmax}\big((\mathbf{z}_l + \mathbf{g}_l)/T\big) \in \Delta^{V-1}, \qquad l = 1, \dots, L. \tag{2}$$

We restrict attention to category positions $\mathcal{P}_k$ (via the category mask) and to the category slice of the vocabulary, writing $\pi_l^{\mathrm{cat}}(a)$ for the relaxed probability that position $l \in \mathcal{P}_k$ takes category $a \in C$. For a constraint (a,b) let $P_l^A = \pi_l^{\mathrm{cat}}(a)$ and $P_l^B = \pi_l^{\mathrm{cat}}(b)$. We define two energies:

$$g^{\mathrm{ord}}_{(a,b)} = \sum_{l \in \mathcal{P}_k} \Big(\textstyle\sum_{l'<l} P^B_{l'}\Big) P^A_l, \tag{3}$$

$$g^{\mathrm{ex}}_{(a,b)} = \mathrm{ReLU}\Big(1 - \textstyle\sum_l P^A_l\Big) + \mathrm{ReLU}\Big(1 - \textstyle\sum_l P^B_l\Big). \tag{4}$$

The *order energy* (3) is the expected number of inverted pairs, i.e. the probability mass of a b appearing *before* an a; it is zero iff no b precedes any a. The *existence energy* (4) penalizes an expected count below one for either constrained category, which prevents the model from trivially satisfying ordering by *omitting* a or b. Both energies are computed only over category positions; POI logits do not enter Eqs. (3)–(4).

4.3 KL-preserving augmented-Lagrangian projection

At a controlled reverse step, we project the backbone logits to reduce constraint violations while staying close to the model's own distribution. Let $\mathbf{y}^{(0)} = \mathbf{z}$ be the detached backbone logits and $\mathbf{y}$ a trainable copy. We solve, per sequence,

$$\begin{aligned} &\min_{\mathbf{y}} \; D_{\mathrm{KL}}\big(\mathrm{softmax}(\mathbf{y}) \,\big\|\, \mathrm{softmax}(\mathbf{y}^{(0)})\big) \\ &\text{s.t. } g^{\mathrm{ord}}_{(a,b)} \le \tau, \quad g^{\mathrm{ex}}_{(a,b)} \le \tau \quad \forall (a,b), \end{aligned} \tag{5}$$

where the KL term anchors the projected distribution to the base model so that fidelity is preserved away from the constrained categories. We optimize the augmented-Lagrangian relaxation of (5):

with $\delta^\bullet = \mathrm{ReLU}(g^\bullet - \tau)$,

$$\mathcal{L}(\mathbf{y}) = w_{\mathrm{kl}}\, D_{\mathrm{KL}} + \sum_{(a,b)} w_{\mathrm{ord}} \Big(\lambda^{\mathrm{ord}}\delta^{\mathrm{ord}} + \tfrac{1}{2}\mu^{\mathrm{ord}}(\delta^{\mathrm{ord}})^2\Big) + \sum_{(a,b)} w_{\mathrm{ex}} \Big(\lambda^{\mathrm{ex}}\delta^{\mathrm{ex}} + \tfrac{1}{2}\mu^{\mathrm{ex}}(\delta^{\mathrm{ex}})^2\Big). \tag{6}$$

An *inner* loop takes projected-gradient steps on $\mathbf{y}$ (SGD with learning rate η, gradient-norm clipping, updates restricted to positions of constrained sequences). An *outer* loop evaluates the *hard* violations $\hat{\delta}$ (from the arg max one-hot assignment, not the relaxation) and updates the dual variables,

$$\lambda \leftarrow \lambda + \mu\,\hat{\delta}, \qquad \mu \leftarrow \min(\alpha\,\mu,\ \mu_{\max}) \text{ if } \hat{\delta} > \tau_{\mathrm{tol}}, \tag{7}$$

stopping early once $\max \hat{\delta} < \tau_{\mathrm{tol}}$. Evaluating the multiplier update on hard violations lets the penalty grow precisely for the sequences and constraints that remain unsatisfied, rather than applying a uniform fixed penalty. Crucially, the projection reweights the sampling distribution over category positions only; it performs *no* POI candidate filtering and imposes *no* hard category→POI consistency mask.

4.4 Distance-distribution regularization

Enforcing category order perturbs the category marginals and, indirectly, the induced spatial statistics. To preserve geographic realism we add an independent distance regularizer that matches the *population* distribution of travel length. From the training split only, we compute the cumulative great-circle (Haversine) route length d of every training sequence with at least two POIs and summarize it as a soft histogram H_{train} over $\log(1+d)$ with fixed Gaussian bins (the reference is content-hashed and cached, never stored in the model). During projection, for each POI position we fix the top-κ candidate POIs under the backbone and draw P routes with a straight-through Gumbel-softmax (exact distance on the forward pass, a bilinear surrogate on the backward pass), forming a differentiable route-length histogram $H_{\mathrm{proj}}(\mathbf{y})$. We add the term

$$w_{\mathrm{dist}}\, D_{\mathrm{KL}}\big(H_{\mathrm{train}} \,\|\, H_{\mathrm{proj}}(\mathbf{y})\big) \tag{8}$$

to the objective (6). Unlike the order/existence energies, this term acts on *POI-position* logits, and it is a batch-level distribution-matching regularizer rather than a per-sample hard constraint; it is applied only to sequences with at least two POIs and uses an independent random stream. Its weight is not tuned per city and is not guaranteed to improve every fidelity metric; we quantify its effect via the No Distance KL ablation (§5).

4.5 Placement in the reverse process and joint sampling

The control is applied in the *late* reverse-diffusion steps—where the predicted categories are informative enough for the energies to be meaningful—specifically during the last K steps and every f steps therein. At each such step, PCDG replaces the backbone category-position logits with their projected values (and, when enabled, the distance-adjusted POI-position logits), and the next discrete state is obtained by the backbone's ordinary joint categorical sampling over the full token sequence, optionally through the category-consistent decoder of §4.6. The temporal branch is unchanged. Because every operation reads and rewrites only the per-step sampling logits of a frozen model, PCDG adds no trainable parameters and requires no retraining, and it reduces to the unconstrained backbone when the projection is disabled. Algorithm pseudocode and default hyperparameters are given in Appendix A and Figure 1.

4.6 Category-consistent POI decoding

The projection of §4.3 acts on category positions, but the backbone emits categories and POIs as separate segments of the same token string and samples every position from its own categorical. Nothing ties the POI emitted at position l to the category sampled at position l. This matters because the constraint metrics are evaluated on the *actual* category of each generated POI, $\mathrm{cat}(\cdot)$ from the POI catalog, not on the category tokens: a sequence whose category tokens are perfectly ordered still violates $\mathbf{M}$ if the POIs realize a different category sequence. On a 128-sequence single-seed diagnostic (not one of the formal runs) the projected backbone left 17.5% of positions category–POI inconsistent; enabling the decoder below drove that rate to 0 and, *without altering a single category token*, cut the strict violation rate from 27.0% to 2.2%—so almost all of those violations came from the POI segment disagreeing with an already correctly ordered category segment.

We therefore decode the two segments *sequentially within one step* rather than independently. Let $\mathbf{g}$ be the Gumbel draw used by the backbone's categorical sampler at this step. We first take the ordinary sample $\hat{x}_l = \arg\max_v(\mathbf{g}+\mathbf{z})_{lv}$, which fixes the realized category at every category position. We then set, at each POI position (the set $\mathcal{P}_{\mathrm{poi}}$ given by the POI mask) the logits of every POI whose catalog category differs from that realized category to $-\infty$ and re-take the arg-max *reusing the same draw* $\mathbf{g}$:

$$\hat{x}_l^{\mathrm{cc}} = \arg\max_v \big(\mathbf{g}+\widetilde{\mathbf{z}}\big)_{lv}, \qquad \widetilde{z}_{lv} = \begin{cases} -\infty & l \in \mathcal{P}_{\mathrm{poi}},\ \mathrm{cat}(v) \neq \hat{c}_l \\ z_{lv} & \text{otherwise,} \end{cases} \tag{9}$$

where $\hat{c}_l$ is the category realized for the event at position l. Three properties follow. (i) Category positions are never rewritten, so the realized category sequence—and hence the ordering the projection just enforced—is preserved exactly. (ii) Exactly one draw is consumed per step, identical in shape and order to the unmodified sampler, so the "off" and "on" runs are paired at matched randomness and no extra global, spatial, projection or distance randomness is used. (iii) Masking uses true $-\infty$ rather than a large finite penalty, which would let a low-scoring legal POI tie with an illegal one. At intermediate reverse steps the POI mask state is still admissible and no restriction is applied when the category is itself masked or no legal POI exists; at the final step only same-category POIs are legal and the sampler raises rather than silently emitting an inconsistent token. The decoder adds no parameters and no logit prior—it is a restriction on realized tokens.

4.7 Terminal same-category geometry refinement

The stages so far decide *which* category and *which* POI identity; the coordinates are then whatever the chosen POIs happen to be, and the spatial marginals drift as a result (§5.4). After the reverse process completes, we therefore re-select each POI *within its own realized category*.

For every POI position we build a candidate set of at most κ_g=32 POIs sharing the realized category: the originally sampled POI, the backbone's highest-scoring candidates, the POIs closest to the generated trajectory's own centroid, and—in the distribution-corrected variant—the most frequently visited POIs of the training reference; duplicates are removed, the set is filled by model score, and ties are broken by catalog id so the construction is deterministic. Scores come from one frozen, untruncated t=0 evaluation of the backbone; reference GPS from the test split never enters. Writing $\mathbf{w}$ for a relaxed assignment over candidates, we minimize

$$D_{\mathrm{KL}}(q \,\|\, \mathbf{w})\, w_{\mathrm{fid}} + w_{\mathrm{d}}\, \mathrm{JSD}\big(H^{\mathrm{dist}}_{\mathrm{train}}, H^{\mathrm{dist}}(\mathbf{w})\big) + w_{\mathrm{r}}\, \mathrm{JSD}\big(H^{\mathrm{rad}}_{\mathrm{train}}, H^{\mathrm{rad}}(\mathbf{w})\big), \tag{10}$$

where the two soft histograms are differentiable in $\mathbf{w}$ and the proposal q places half of its mass on the original POI and spreads the other half over the candidates by model score, so the fidelity term penalizes drifting away from the backbone's own preference. We optimize Eq. (10) with P_g=8 straight-through relaxed paths, temperature 1, Adam at learning rate 0.05, for a prequalified budget of 200 steps, refreshing an independent geometry noise stream every 20 steps. The output is a *hard* assignment: among the original assignment and the 8 optimized paths we keep the arg-min of the same training-side objective. Empty, single-point and single-candidate positions, and batches in which the base projection never ran, are skipped.

Constraint neutrality. Because the refinement substitutes only POIs of the same realized category, together with their catalog coordinates, and leaves category tokens, the realized category sequence, event times, sequence lengths, event order and all conditions unchanged *per event*, every constraint-relevant statistic is invariant by construction: $\mathrm{OVR}_{\mathrm{strict}}$, $\mathrm{OVR}_{\mathrm{skip}}$, pair and category coverage, Unsat, *Category*, *CategoryTransition* and the category–POI mismatch rate. We verify this numerically on every test run, requiring agreement within 10^{-12} rather than assuming it. The refinement can therefore only move the spatial and popularity statistics—which is precisely why it can repay the fidelity cost of the constraint stages without giving any of their gains back.

4.8 Distribution correction

Three scalars govern the balance between constraint strength and distributional fidelity, all referenced to the *training* reference pool only. First, $\rho \in (0, 1]$ scales the constraint term of Eq. (6), so the projection objective becomes

$$\begin{aligned} w_{\mathrm{kl}}\, D_{\mathrm{KL}} + \rho \sum_{(a,b)} \mathrm{AL}\big(g^{\mathrm{ord}}, g^{\mathrm{ex}}; \lambda^{\bullet}, \mu^{\bullet}\big) \\ + w_{\mathrm{dist}}\, D_{\mathrm{KL}}\big(H_{\mathrm{train}} \| H_{\mathrm{proj}}\big) + \lambda\, \mathcal{L}_{\mathrm{cat}}. \end{aligned} \tag{11}$$

Second, $\lambda \geq 0$ weights a category-side term

$$\mathcal{L}_{\mathrm{cat}} = \tfrac{1}{2}\big(\mathrm{JSD}_{\mathrm{hour\times cat}} + \mathrm{JSD}_{\mathrm{transition}}\big), \tag{12}$$

computed from *expected* counts under the relaxed assignment—no Monte Carlo draws, so no additional randomness is consumed—with self-transitions counted, sequence boundaries excluded, and reference rows without support contributing zero. Third, $\beta \geq 0$ adds to Eq. (10) a POI-side term $\frac{1}{2}(\mathrm{JSD}_{\mathrm{top1000}} + \mathrm{JSD}_{\mathrm{DailyLoc}})$, where the first is an *identity-aligned* top-1000 visitation divergence (the generated and reference top-1000 sets are unioned by POI id before normalization, not compared as sorted popularity curves) and the second a soft expected-*DailyLoc* divergence. Spatial optimization results are cached independently of β, so the three β settings of a batch are compared at identical upstream randomness.

The configuration used throughout is ρ=0.5, λ=0, β=1, selected on a training-side calibration grid (Appendix B). We retain λ in the formulation because it is the natural category-side counterpart of β; that the selected configuration switches it *off* is an empirical finding about where the fidelity cost lives (§5.4), not a modelling assumption.

5 Experiments

We evaluate whether PCDG enforces category precedence while preserving fidelity, how it compares to retraining- and post-processing-based alternatives, and how each component contributes.

5.1 Setup

Datasets. We use check-in data from two cities, **New York** and **Istanbul**, under an out-of-distribution (OOD) split that holds out partial-order configurations unseen during training (2,108 and 4,914 held-out reference sequences, respectively). For each reference we derive a sample-level partial-order matrix $\mathbf{M}$ from the category first/last positions (§3). The base generators are trained once per city and **kept frozen** for all methods; PCDG and all sampling-time baselines reuse the identical frozen weights. [TODO: confirm final train/test counts and OOD definition from distance-v2 manifest]

Protocol. Unless noted, the constraint projection uses order weight w_{ord}=1, existence weight w_{ex}=5, token-KL weight w_{kl}=1, and distance-KL weight w_{dist}=1 (8 routes, top-32 candidates, 32 bins), with Gumbel temperature 3, applied over the last 40 reverse steps every 4 steps without early stopping. No per-city hyper-parameter search is performed. We report the mean ± sample standard deviation over **three sampling seeds** at a fixed checkpoint; the reported variability reflects sampling stochasticity, not training variability.

Baselines. We compare four external methods on the same frozen backbone unless stated: **JointGen** (unconstrained backbone); **PostSwap** (generate, then stably reorder categories to satisfy $\mathbf{M}$); **EnergyGuide** (gradient guidance by the constraint-violation energy during sampling [33]); and **PO-CFG** (a *retrained* classifier-free-guidance model conditioned on the partial order, trained for 1000 epochs; guidance scale 1). PO-CFG is the only method that requires retraining. We report three variants of our own stack: PCDG-P (projection only), PCDG-PD (adding category-consistent POI decoding), and the full PCDG (adding distribution-aware geometry refinement).

Metrics. *Fidelity* uses the Jensen–Shannon divergence (JSD, lower is better) of six marginals (evaluation protocol v2): travel *Distance*, *Radius* of gyration, *CategoryTransition* (adjacent directed category pairs), *DailyLoc* (daily location count), hourly *Category* usage, and a global rank statistic *G-RANK*. *Constraint satisfaction* (percentages) uses OVR_{strict} and OVR_{skip} (ordering-violation rates), *Pair coverage* and *Category coverage* (fraction of required relations/categories realized, higher is better), and *Unsat* (fraction of sequences violating any constraint, lower is better). [TODO: add one-line formal definitions in Appendix D]

5.2 Main results

Table 1 reports the method comparison on both cities.

Constraint satisfaction. The full PCDG reduces the strict ordering-violation rate from 92.89% to 4.04% on Istanbul and from 89.45% to 11.28% on New York (−88.85 and −78.17 percentage points; −95.7% and −87.4% relative), lowers *Unsat* from 95.57% to 7.29% and from 93.64% to 15.86%, and raises pairwise ordering coverage by 6.5× and 4.2× respectively. It exceeds the *retrained* PO-CFG on all five constraint metrics in both cities by a wide margin (63.45% → 4.04% and 41.91% → 11.28% strict violations), despite training nothing. PostSwap drives $\mathrm{OVR}_{\mathrm{skip}}$ to 0 by construction, but because it cannot introduce a missing category its coverage is identical to JointGen's—showing that skip=0 must not be read as constraint satisfaction. EnergyGuide, a single gradient correction without dual updates, barely moves any constraint metric.

Fidelity. The projection stage alone reproduces the trade-off reported for constrained sampling: PCDG-P improves *CategoryTransition*, *DailyLoc* and *Category* on Istanbul but degrades *Distance* (0.1128 → 0.1314) and *Radius* (0.0226 → 0.0516). The full stack removes this cost. On Istanbul *all six* divergences fall below the unconstrained backbone (e.g. Distance −65.0%, Radius −26.5%, DailyLoc −39.9%), and five of six fall below PO-CFG, which was the best raw-fidelity method in the projection-only comparison (G-RANK is effectively tied, 0.1798 vs. 0.1794). On New York four of six improve over the backbone (Distance −76.0%, Radius −66.2%, DailyLoc −26.4%, Category −4.9%).

Residuals we do not hide. On New York, *CategoryTransition* (0.0070 → 0.0183) and *G-RANK* (0.1612 → 0.1914) remain above the unconstrained backbone for every constrained variant of ours, and PO-CFG remains the best method on exactly those two plus *Category*. Enforcing an ordering the data does not exhibit necessarily rewrites adjacent category pairs, so some *CategoryTransition* loss is intrinsic to the task rather than to our mechanism; the *G-RANK* gap is a POI-popularity effect that the distribution correction reduces but does not close on this city (§5.4). All comparisons are between three-seed means; we run no significance tests.

5.3 Ablation of the projection stage

Tables 2 and 3 (in Appendix C, for space) isolate each component by disabling it from the projection stage: No Projection (the backbone), No Existence, No Order, No KL (both KL terms off), No Distance KL (only the distance term off), Fixed Multipliers (no dual update, Eq. (7)), No Gumbel (deterministic relaxation), and Full (= PCDG-P).

Findings. The two energies have cleanly separated roles. Removing the *existence* energy collapses pair coverage to 7.34% (Istanbul) and 10.95% (New York)—*below* the unconstrained backbone's 14.63% and 21.13%—while $\mathrm{OVR}_{\mathrm{skip}}$ drops to 7.67% and 18.04%: without it, the model satisfies ordering by *omitting* the constrained categories, which is exactly the degeneracy Eq. (4) is designed to block. Removing the *order* energy keeps coverage high (72.53% / 69.92%) but inflates $\mathrm{OVR}_{\mathrm{strict}}$ from 32.80% / 38.51% to 63.50% / 65.43%. Freezing the multipliers costs 9.90 and 19.94 percentage points of $\mathrm{OVR}_{\mathrm{strict}}$, confirming that evaluating the dual update on *hard* violations matters more than the penalty magnitude alone. The *distance-KL* term improves both spatial metrics in both cities (Istanbul Distance 0.1441 → 0.1314, Radius 0.0582 → 0.0516; New York 0.2484 → 0.2210 and 0.1430 → 0.1265) at no measurable constraint cost. The *token-KL* and *Gumbel* terms have small and not uniformly positive marginal effects at this projection budget—removing the Gumbel noise even lowers New York *Distance* to 0.2055—and we report this rather than claiming a benefit for every component.

5.4 Decomposing the fidelity–constraint trade-off

Figure 2 compares the stages in the Unsat–Distance plane; Table 4 in Appendix C gives all stage-wise means.

Where fidelity is lost. Two stages buy constraint satisfaction and both charge for it on the *spatial* axis. The projection (b) raises Istanbul *Radius* from 0.0226 to 0.0516, and category-consistent decoding (c)—which delivers the largest single constraint jump, 32.80% → 3.72% and 38.51% → 9.52% strict violations—raises it further to 0.0892 (0.1265 to 0.1681 on New York). Neither stage degrades the temporal or category-usage marginals; the damage is concentrated in *Distance* and *Radius*.

Where it is recovered. The same-category geometry refinement (d) is constraint-neutral by construction—it swaps only POIs of the same *realized* category together with their catalog coordinates, so every constraint metric is bit-identical to (c) to within 10^{-12}—and it repays the entire spatial cost and more: Istanbul *Distance* 0.1539 → 0.0372 and *Radius* 0.0892 → 0.0159, New York 0.1989 → 0.0610 and 0.1681 → 0.0382. This is the central empirical point: the trade-off is not intrinsic to constrained sampling, it is an artifact of selecting POIs and coordinates without reference to the population spatial distribution. The refinement does, however, concentrate mass on its candidate pools and so degrades the POI-popularity statistic on New York (*G-RANK* 0.1914 → 0.2069). Stage (e) adds the distribution-correction terms and a halved constraint weight, which repairs exactly that statistic (0.2069 → 0.1914; Istanbul 0.1835 → 0.1798) and improves *DailyLoc* in both cities, at the price of a small constraint regression (3.72 → 4.04 and 9.52 → 11.28) and a slightly higher *Distance*.

The loss is spatial, not categorical. The configuration selected from the $\rho \times \lambda \times \beta$ grid (Appendix B) is ρ=0.5, λ=0, β=1: it halves the constraint penalty, applies the POI-distribution term at full weight, and switches the category-side distribution term *off* entirely. Combined with the stage-wise pattern above, this says the residual fidelity cost of partial-order control in this generator lives almost entirely on the spatial/POI side, and that a category-side distribution regularizer is not needed to correct it.

Selection protocol. The 18 configurations were ranked by a fixed rule—maximum, then mean, of the twelve per-city JSD/JointGen ratios—computed on a *training-side* calibration split, and the top-ranked configuration was then evaluated once on the held-out OOD test set. No test metric entered candidate selection, hyper-parameter choice, candidate construction, or the refinement objective. We note explicitly that this calibration split is an engineering calibration rather than an independent validation set (the frozen backbone was trained on those records), and that of the two configurations which reached three-seed confirmation neither cleared all pre-specified

Table 1: Method comparison (evaluation v2): fidelity JSDs (upper panel) and constraint metrics in percent (lower panel). Cells are mean ± sample SD over three sampling seeds at fixed checkpoints, not confidence intervals. Bold marks the best unrounded city-wise mean, not significance. PCDG-P: projection; PCDG-PD: adds decoding; PCDG: full stack. New York CategoryTransition and G-RANK remain above JointGen.

Istanbul						
Method	Distance ↓	Radius ↓	CategoryTransition ↓	DailyLoc ↓	Category ↓	G-RANK ↓
JointGen	0.1128 ± 0.0087	0.0226 ± 0.0017	0.0403 ± 0.0001	0.0414 ± 0.0009	0.0649 ± 0.0003	0.1922 ± 0.0022
PostSwap	0.0949 ± 0.0127	0.0226 ± 0.0017	0.0361 ± 0.0005	0.0414 ± 0.0009	0.0647 ± 0.0018	0.1922 ± 0.0022
EnergyGuide	0.1156 ± 0.0052	0.0222 ± 0.0025	0.0424 ± 0.0009	0.0392 ± 0.0020	0.0643 ± 0.0016	0.1921 ± 0.0004
PO-CFG (s=1)	0.0890 ± 0.0050	0.0263 ± 0.0020	0.0365 ± 0.0008	0.0444 ± 0.0022	0.0630 ± 0.0007	**0.1794 ± 0.0026**
PCDG-P (projection)	0.1314 ± 0.0035	0.0516 ± 0.0022	**0.0162 ± 0.0006**	0.0368 ± 0.0014	0.0446 ± 0.0020	0.1820 ± 0.0016
PCDG-PD (+decoding)	0.1539 ± 0.0028	0.0892 ± 0.0024	0.0238 ± 0.0011	0.0297 ± 0.0012	0.0423 ± 0.0009	0.1813 ± 0.0018
PCDG (full)	**0.0395 ± 0.0036**	**0.0166 ± 0.0007**	0.0235 ± 0.0011	**0.0249 ± 0.0009**	**0.0423 ± 0.0018**	0.1798 ± 0.0016
New York						
Method	Distance ↓	Radius ↓	CategoryTransition ↓	DailyLoc ↓	Category ↓	G-RANK ↓
JointGen	0.2904 ± 0.0156	0.1104 ± 0.0071	0.0070 ± 0.0008	0.0409 ± 0.0035	0.0184 ± 0.0002	0.1612 ± 0.0013
PostSwap	0.2276 ± 0.0448	0.1104 ± 0.0071	0.0070 ± 0.0007	0.0409 ± 0.0035	0.0168 ± 0.0006	0.1612 ± 0.0013
EnergyGuide	0.2943 ± 0.0060	0.1112 ± 0.0061	0.0071 ± 0.0011	0.0411 ± 0.0029	0.0183 ± 0.0003	0.1622 ± 0.0009
PO-CFG (s=1)	0.1228 ± 0.0160	0.0823 ± 0.0029	**0.0053 ± 0.0018**	0.0499 ± 0.0008	**0.0121 ± 0.0007**	**0.1525 ± 0.0025**
PCDG-P (projection)	0.2210 ± 0.0224	0.1265 ± 0.0051	0.0132 ± 0.0004	0.0351 ± 0.0035	0.0173 ± 0.0013	0.1942 ± 0.0041
PCDG-PD (+decoding)	0.1989 ± 0.0198	0.1681 ± 0.0042	0.0182 ± 0.0012	0.0335 ± 0.0022	0.0178 ± 0.0012	0.1914 ± 0.0059
PCDG (full)	**0.0698 ± 0.0107**	**0.0373 ± 0.0042**	0.0183 ± 0.0005	**0.0301 ± 0.0010**	0.0175 ± 0.0010	0.1914 ± 0.0060

Istanbul					
Method	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
JointGen	92.89 ± 0.08	48.05 ± 0.74	14.63 ± 0.19	43.19 ± 0.20	95.57 ± 0.15
PostSwap	86.10 ± 0.15	**0.00 ± 0.00**	14.63 ± 0.19	43.19 ± 0.20	85.37 ± 0.19
EnergyGuide	90.66 ± 0.06	44.76 ± 0.58	19.17 ± 0.43	47.06 ± 0.25	92.89 ± 0.16
PO-CFG (s=1)	63.45 ± 0.23	19.09 ± 0.54	43.72 ± 0.35	66.00 ± 0.23	69.07 ± 0.45
PCDG-P (projection)	32.80 ± 0.50	9.12 ± 0.10	70.99 ± 0.08	85.36 ± 0.15	36.86 ± 0.37
PCDG-PD (+decoding)	**3.72 ± 0.13**	1.76 ± 0.10	**94.74 ± 0.16**	**97.81 ± 0.07**	**6.90 ± 0.18**
PCDG (full)	4.04 ± 0.20	2.02 ± 0.15	94.68 ± 0.15	97.75 ± 0.10	7.29 ± 0.20
New York					
Method	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
JointGen	89.45 ± 0.20	52.92 ± 0.86	21.13 ± 0.15	46.75 ± 0.12	93.64 ± 0.26
PostSwap	78.22 ± 0.48	**0.00 ± 0.00**	21.13 ± 0.15	46.75 ± 0.12	78.87 ± 0.15
EnergyGuide	89.17 ± 0.14	51.59 ± 0.88	21.17 ± 0.19	47.14 ± 0.14	93.20 ± 0.39
PO-CFG (s=1)	41.91 ± 1.37	8.97 ± 0.88	59.92 ± 1.58	74.99 ± 1.08	49.48 ± 1.20
PCDG-P (projection)	38.51 ± 0.95	14.11 ± 1.22	67.60 ± 1.86	81.54 ± 0.70	42.42 ± 1.84
PCDG-PD (+decoding)	**9.52 ± 1.04**	3.47 ± 0.64	**88.67 ± 1.74**	**93.92 ± 0.68**	**14.09 ± 2.11**
PCDG (full)	11.28 ± 1.05	4.50 ± 0.70	87.87 ± 1.59	93.46 ± 0.63	15.86 ± 1.90

calibration thresholds; we report the resulting held-out test numbers as-is rather than re-opening the search.

5.5 Cost and reproducibility

PCDG adds only sampling-time projection iterations to a frozen generator and trains nothing, whereas PO-CFG requires a full retraining per city. We report wall-clock sampling overhead and the number of projection updates in Appendix E, and we release code, constraint matrices, and evaluation scripts to reproduce every table. [TODO: insert sampling-time overhead numbers from distance-v2 diagnostics]

6 Limitations

The study covers two cities and three sampling seeds at fixed checkpoints, not variation across training runs or a significance test. Istanbul's historical backbone used a different training batch size from New York, and its full training provenance cannot be reconstructed. PO-CFG is evaluated at scale 1, not an extrapolation-tuned optimum. The fixed distance-KL weight is shared across cities; its batch-level objective gives no per-sequence spatial guarantee.

PCDG reduces constraint violations without certifying feasibility. The decoder uses hard POI-support masking but permits a catalog fallback for unsupported final categories (§4.6). New York CategoryTransition (0.0183 vs. 0.0070) and G-RANK (0.1914 vs. 0.1612) remain above JointGen. Hyperparameters were selected on training-side calibration records already seen by the backbone; neither three-seed-confirmed candidate passed all distribution thresholds. Test metrics never entered that selection.

The (d)→(e) comparison changes ρ, β, and candidate-pool composition jointly, precluding single-factor attribution. Stage (d) transfers an already-fixed geometry configuration rather than establishing a separately tuned generalization result. Complex partial-order DAGs, conflict detection, adaptive scheduling, training-time order

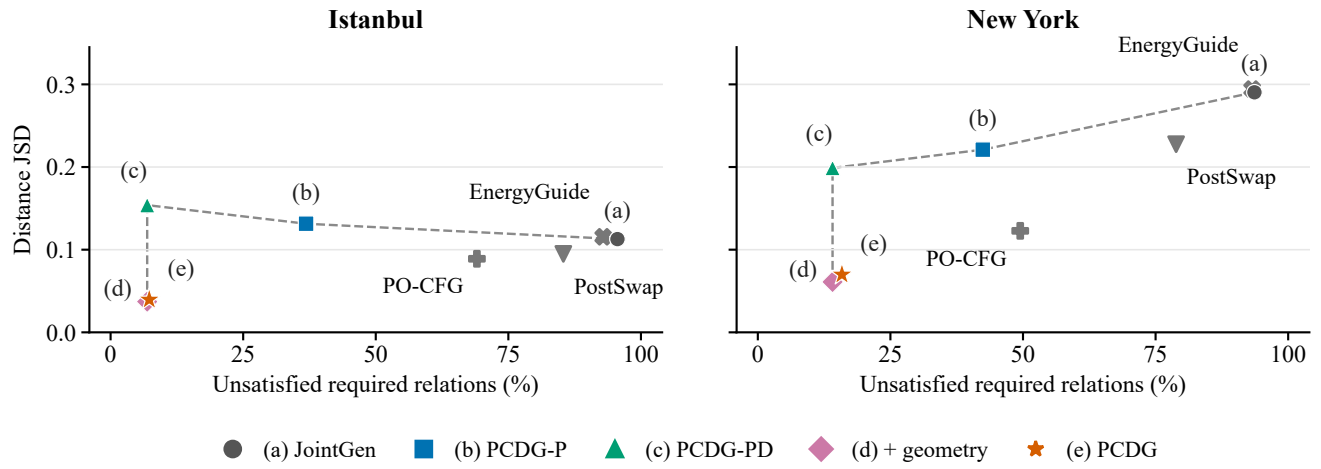


Figure 2: Unsat–Distance trade-off using the evaluation-v2 three-seed means in Tables 1 and 4; lower is better on both axes. Dashed paths connect (a)–(e); shared method/stage points are plotted once. Geometry (c)→(d) lowers Distance JSD without changing Unsat. Transition (d)→(e) changes three factors jointly, not a single regularizer.

losses, and downstream recommendation utility remain open. Synthetic output has no formal privacy guarantee here; distributional similarity alone does not establish safe data release.

7 Conclusion

PCDG provides sampling-time partial-order control for a frozen POI check-in generator. Its order and existence energies control category logits, category-consistent decoding reduces disagreement with POI identities, and same-category geometry refinement improves spatial fidelity while preserving constraint-relevant fields. On two cities, the full stack improves all five constraint metrics over the retrained PO-CFG baseline. It improves all six fidelity JSDs over JointGen on Istanbul and four on New York. The stage-wise results show that spatial distortion can be reduced independently of constraint satisfaction in this setting, while the remaining category-transition and popularity gaps bound the scope of that separation.

A Algorithm and Default Hyper-parameters

Algorithm 1 summarizes one projected reverse step. Defaults: w_{ord}=1, w_{ex}=5, w_{kl}=1, w_{dist}=1; Gumbel temperature T=3; outer iterations 10, inner iterations 50; SGD learning rate η, gradient clip 10; multiplier growth α=2, $\mu_{\max}$=10^3, tolerance τ_{tol}=0.25; distance routes P=8, candidates κ=32, bins 32; applied over the last 40 reverse steps every 4 steps. Geometry refinement (§4.7): w_{d}=1, w_{r}=2, w_{fid}=0.1, P_g=8 paths, κ_g=32 candidates, temperature 1, Adam learning rate 0.05, 200 steps, geometry noise refreshed every 20 steps. Distribution correction (§4.8): ρ=0.5, λ=0, β=1. The geometry step budget was fixed by a speed qualification (median total spatial-sampling time within 1.10× of the unrefined backbone) before any quality comparison, and the weights by the grid of Appendix B. Algorithm 1 covers one projected reverse step only; the decoder of §4.6 acts inside its final line, and the refinement of §4.7 acts once after the reverse process terminates. [TODO: confirm η and step schedule against the distance-v2 manifest before camera-ready.]

Algorithm 1 PCDG projected reverse step (one controlled step s)

Require: backbone logits $\mathbf{z}$; constraints $\{(a, b)\}$ with $\mathbf{W}_A, \mathbf{W}_B$; masks; distance reference H_{train}

1: $\mathbf{y} \leftarrow \mathbf{z}$ (detached); initialize λ, μ for active rows
2: **for** outer = 1 to T_{out} **do**
3: **for** inner = 1 to T_{in} **do**
4: relax $\mathbf{y}$ to $\boldsymbol{\pi}$; compute $g^{\text{ord}}, g^{\text{ex}}$ on category positions ▷ Eq. (3)–(4)
5: $\mathcal{L} \leftarrow w_{\text{kl}} D_{\text{KL}} + \text{AL}(g^{\text{ord}}, g^{\text{ex}}; \lambda, \mu) + w_{\text{dist}} D_{\text{KL}}(H_{\text{train}} \| H_{\text{proj}})$
6: SGD step on $\mathbf{y}$ (clip; restrict to active rows)
7: **end for**
8: evaluate hard violations $\hat{\delta}$; update λ, μ ▷ Eq. (7); early stop if satisfied
9: **end for**
10: **return** joint categorical sample from projected $\mathbf{y}$, decoded category-first ▷ Eq. (9)

B Distribution-Correction Grid and Selection

The distribution-correction stage has three weights. ρ scales the existing constraint loss inside the augmented-Lagrangian objective (Eq. (6)); λ weights a category-side distribution term—the mean of an hour–category JSD and a conditional category-transition JSD against the training reference pool; and β weights a POI-side distribution term—the mean of an identity-aligned top-1000 visitation JSD and a soft expected-*DailyLoc* JSD—inside the same-category geometry refinement objective. We screened the full grid $\rho \in \{0.25, 0.5, 1\} \times \lambda \in \{0, 1\} \times \beta \in \{0, 0.1, 1\}$ (18 configurations) on a single seed, retained at most two jointly feasible configurations

per city, and confirmed those over three seeds, all on a training-side calibration split with 512 conditions per city and seed. Configurations were ranked by the maximum, then the mean, of the twelve per-city JSD/JointGen ratios, with the parameter tuple as the final tie-break. The two configurations that reached confirmation were $(\rho, \lambda, \beta) = (0.5, 0, 1)$ and $(0.5, 0, 0.1)$; the first was selected and is the configuration reported throughout. Spatial results are cached by $(city, pool, seed, \rho, \lambda)$ and reused across the three β refinements of each batch, so β varies at fixed upstream randomness. No test-set metric participated in screening, confirmation, ranking, candidate construction, or the refinement objective. As stated in §6, neither confirmed configuration cleared all pre-specified calibration thresholds: both met the *Distance*, *Radius* and *DailyLoc* ceilings in both cities but not the *CategoryTransition*, *Category* and *G-RANK* ceilings on that calibration split. On the held-out test set the selected configuration then met all eleven gates on Istanbul and nine of eleven on New York. [TODO: Appendix table: all 18 screened configurations with their twelve calibration ratios, from screen-selection.json.]

C Projection and Stage-wise Tables

Tables 2 and 3 are referenced from §5.3 and placed here for space.

D Metric Definitions

[TODO: one-line formal definitions of OVR_strict, OVR_skip, Pair/Category coverage, Unsat, and the six eval-v2 JSD marginals; copy from the unified evaluation module.]

Two definitions that differ from common usage. Radius is computed as $\sqrt{\text{mean}(d(\cdot, \bar{c}))}$, the square root of the mean distance to the arithmetic centroid, *not* the standard root-mean-square radius of gyration; it is therefore comparable across the rows of our tables but not directly against differently defined radii in prior work. *G-RANK* compares the visitation distributions of the top-1000 most frequent POIs; because many POIs tie at the truncation frequency, the membership of that top-1000 set is implementation-dependent under an unstable sort. All tables in this paper use the archived evaluation protocol (v2). We additionally recomputed a deterministic variant that sorts by count descending then POI id ascending and aligns the generated and reference sets by identity before normalization. The effect on *G-RANK* is at most 0.0032 in absolute value and changes no ordering or conclusion: on Istanbul the full stack remains below the backbone (0.1766 vs. 0.1892) and on New York it remains above it (0.1912 vs. 0.1628). All other statistics agree with the archived values to within 10^{-12}.

E Cost

[TODO: sampling wall-clock overhead vs. JointGen, and PO-CFG retraining cost (per-city GPU-hours), from distance-v2 diagnostics.]

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Table 2: Ablation of the projection stage: distributional similarity (JSD ↓, evaluation v2). Full here is PCDG-P, i.e. the projection stage alone.

Istanbul						
Variant	Distance ↓	Radius ↓	CategoryTransition ↓	DailyLoc ↓	Category ↓	G-RANK ↓
No Projection	0.1128 ± 0.0087	0.0226 ± 0.0017	0.0403 ± 0.0001	0.0414 ± 0.0009	0.0649 ± 0.0003	0.1922 ± 0.0022
No Existence	**0.1084 ± 0.0114**	**0.0215 ± 0.0027**	0.0389 ± 0.0005	0.0412 ± 0.0021	0.0661 ± 0.0009	0.1924 ± 0.0028
No Order	0.1176 ± 0.0023	0.0486 ± 0.0035	0.0227 ± 0.0008	**0.0358 ± 0.0006**	0.0439 ± 0.0018	**0.1770 ± 0.0040**
No KL	0.1406 ± 0.0117	0.0574 ± 0.0028	0.0164 ± 0.0006	0.0361 ± 0.0012	**0.0429 ± 0.0012**	0.1817 ± 0.0024
Fixed Multipliers	0.1296 ± 0.0093	0.0526 ± 0.0045	0.0185 ± 0.0005	0.0361 ± 0.0009	0.0457 ± 0.0025	0.1841 ± 0.0020
No Gumbel Noise	0.1315 ± 0.0091	0.0524 ± 0.0023	**0.0160 ± 0.0010**	0.0370 ± 0.0005	0.0439 ± 0.0023	0.1831 ± 0.0027
No Distance KL	0.1441 ± 0.0073	0.0582 ± 0.0025	0.0162 ± 0.0010	0.0366 ± 0.0008	0.0437 ± 0.0023	0.1828 ± 0.0026
Full (=PCDG-P)	0.1314 ± 0.0035	0.0516 ± 0.0022	0.0162 ± 0.0006	0.0368 ± 0.0014	0.0446 ± 0.0020	0.1820 ± 0.0016
New York						
Variant	Distance ↓	Radius ↓	CategoryTransition ↓	DailyLoc ↓	Category ↓	G-RANK ↓
No Projection	0.2904 ± 0.0156	0.1104 ± 0.0071	**0.0070 ± 0.0008**	0.0409 ± 0.0035	0.0184 ± 0.0002	0.1612 ± 0.0013
No Existence	0.2439 ± 0.0277	**0.1004 ± 0.0053**	0.0075 ± 0.0008	0.0393 ± 0.0044	0.0189 ± 0.0004	**0.1587 ± 0.0006**
No Order	0.2175 ± 0.0284	0.1196 ± 0.0108	0.0138 ± 0.0003	0.0352 ± 0.0029	0.0177 ± 0.0003	0.1860 ± 0.0048
No KL	0.2415 ± 0.0239	0.1421 ± 0.0046	0.0125 ± 0.0006	0.0352 ± 0.0041	0.0173 ± 0.0013	0.1939 ± 0.0064
Fixed Multipliers	0.2402 ± 0.0487	0.1218 ± 0.0018	0.0127 ± 0.0005	0.0353 ± 0.0041	0.0175 ± 0.0006	0.1891 ± 0.0023
No Gumbel Noise	**0.2055 ± 0.0205**	0.1248 ± 0.0056	0.0130 ± 0.0010	0.0360 ± 0.0036	0.0173 ± 0.0003	0.1918 ± 0.0041
No Distance KL	0.2484 ± 0.0267	0.1430 ± 0.0055	0.0134 ± 0.0011	0.0355 ± 0.0031	**0.0172 ± 0.0013**	0.1937 ± 0.0082
Full (=PCDG-P)	0.2210 ± 0.0224	0.1265 ± 0.0051	0.0132 ± 0.0004	**0.0351 ± 0.0035**	0.0173 ± 0.0013	0.1942 ± 0.0041

Mean ± sample standard deviation over three sampling seeds. Bold denotes the best unrounded mean within each city, not statistical significance. No KL disables both the token-KL anchor and the distance-KL term; No Distance KL disables only the latter. Removing the KL or Gumbel terms does not uniformly degrade fidelity, which we report rather than claiming a benefit for every component.

Table 3: Ablation of the projection stage: constraint satisfaction (%, evaluation v2). The existence energy drives coverage, the order energy drives strict ordering, and the adaptive multipliers beat a fixed penalty.

Istanbul					
Variant	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
No Projection	92.89 ± 0.08	48.05 ± 0.74	14.63 ± 0.19	43.19 ± 0.20	95.57 ± 0.15
No Existence	93.67 ± 0.15	**7.67 ± 0.56**	7.34 ± 0.12	36.92 ± 0.19	93.58 ± 0.17
No Order	63.50 ± 0.76	50.34 ± 0.73	**72.53 ± 0.70**	**86.08 ± 0.17**	70.70 ± 0.73
No KL	32.85 ± 0.52	9.20 ± 0.29	71.11 ± 0.08	85.40 ± 0.16	36.75 ± 0.57
Fixed Multipliers	42.70 ± 0.51	20.97 ± 0.90	71.25 ± 0.89	85.00 ± 0.42	45.64 ± 0.58
No Gumbel Noise	**32.75 ± 0.39**	9.13 ± 0.22	70.97 ± 0.34	85.38 ± 0.19	**36.75 ± 0.25**
No Distance KL	32.96 ± 0.46	9.10 ± 0.16	70.92 ± 0.23	85.29 ± 0.14	36.83 ± 0.24
Full (=PCDG-P)	32.80 ± 0.50	9.12 ± 0.10	70.99 ± 0.08	85.36 ± 0.15	36.86 ± 0.37
New York					
Variant	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
No Projection	89.45 ± 0.20	52.92 ± 0.86	21.13 ± 0.15	46.75 ± 0.12	93.64 ± 0.26
No Existence	90.38 ± 0.47	18.04 ± 1.26	10.95 ± 0.45	39.47 ± 0.12	91.80 ± 0.40
No Order	65.43 ± 0.12	51.24 ± 0.30	**69.92 ± 0.74**	**82.84 ± 0.40**	72.86 ± 0.94
No KL	**38.24 ± 0.96**	13.98 ± 0.82	67.67 ± 1.56	81.57 ± 0.72	**42.40 ± 1.48**
Fixed Multipliers	58.45 ± 0.66	35.14 ± 0.60	63.79 ± 0.76	78.37 ± 0.35	60.67 ± 0.49
No Gumbel Noise	39.11 ± 1.53	13.93 ± 1.34	66.41 ± 1.32	80.95 ± 0.65	43.91 ± 1.59
No Distance KL	38.48 ± 1.04	**13.76 ± 0.72**	67.33 ± 1.86	81.38 ± 0.81	42.77 ± 1.79
Full (=PCDG-P)	38.51 ± 0.95	14.11 ± 1.22	67.60 ± 1.86	81.54 ± 0.70	42.42 ± 1.84

Mean ± sample standard deviation over three sampling seeds. Bold denotes the best unrounded mean within each city. Removing the existence energy collapses coverage *below* the unconstrained backbone, confirming that ordering can otherwise be satisfied by omitting the constrained categories.

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Table 4: Stage-wise comparison, evaluation-v2 three-seed means. SDs for (a), (b), (c), and (e) are in Table 1. Geometry (d) preserves (c)'s constraints within 10^{-12} and uses a previously fixed configuration. Transition (d)→(e) jointly changes ρ $(1 \to 0.5)$, β $(0 \to 1)$, and the same-category candidate pool; it does not isolate a single factor. The calibration grid (Appendix B) holds the pool construction fixed.

	Distributional similarity (JSD ↓)						Constraints (%)		
Stage	Distance	Radius	CatTrans	DailyLoc	Category	G-RANK	OVR_str. ↓	Pair cov. ↑	Unsat ↓
Istanbul									
(a) JointGen (no control)	0.1128	0.0226	0.0403	0.0414	0.0649	0.1922	92.89	14.63	95.57
(b) + order/existence projection	0.1314	0.0516	0.0162	0.0368	0.0446	0.1820	32.80	70.99	36.86
(c) + category-consistent decoding	0.1539	0.0892	0.0238	0.0297	0.0423	0.1813	3.72	94.74	6.90
(d) + geometry refinement	0.0372	0.0159	0.0238	0.0284	0.0423	0.1835	3.72	94.74	6.90
(e) + distribution correction (full)	0.0395	0.0166	0.0235	0.0249	0.0423	0.1798	4.04	94.68	7.29
New York									
(a) JointGen (no control)	0.2904	0.1104	0.0070	0.0409	0.0184	0.1612	89.45	21.13	93.64
(b) + order/existence projection	0.2210	0.1265	0.0132	0.0351	0.0173	0.1942	38.51	67.60	42.42
(c) + category-consistent decoding	0.1989	0.1681	0.0182	0.0335	0.0178	0.1914	9.52	88.67	14.09
(d) + geometry refinement	0.0610	0.0382	0.0182	0.0335	0.0178	0.2069	9.52	88.67	14.09
(e) + distribution correction (full)	0.0698	0.0373	0.0183	0.0301	0.0175	0.1914	11.28	87.87	15.86

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